Automating Electrical Grid Asset Inspections using Deep Vision: Evaluating the Impact of Annotation Detail Level

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***Abstract*—Automating visual inspections improves efficiency and consistency by addressing the limitations of manual checks, which require skilled labor and are prone to errors due to fatigue and cognitive bias. Automated systems reduce costs but come with their own expenses and challenges. Deep learning models have enhanced computer vision generalization capabilities, being robust to label noise with large datasets. However, clean labels yield better performance than noisy ones. Recent research often focuses on model-centric improvements, neglecting the importance of high-quality data, especially in industries with limited datasets. This study analyzes the impact of annotation detail and quality on model performance, fine-tuning several YOLOv11 models, in a data-centric approach of improving the quality of a high-resolution HV insulator image set, that is evaluated based on three vectors: annotations consistency (mitigating noise with strict labeling); spatial granularity (large box, small box, and polygon); and categorical definition (varying the number of annotated classes). The experimental results evaluate performance across different data quality levels, demonstrating improved test accuracy through enhanced label precision. Furthermore, the study concluded that higher data quality reduces the need for large quantities of annotated data. The conclusions of this work are important to define updated guidelines for optimal labeling effort management.**

***Index Terms*—ASSET INSPECTION, HV INSULATOR, DEEP VISION, DATA-CENTRIC AI, LABEL NOISE**

I. Introduction

Electrical grid increasing reliability is driven by the rising dependency on electrical power. The electrical power growing demand, combined with the integration of renewable energy sources and distributed generation, not only increases the electrical grid management complexity but also requires a significant expansion of the grid in the near future, intensifying the demand for efficient maintenance strategies [1].

The IEEE Power and Energy Society acknowledges the importance of using advanced asset management solutions and automated processes and techniques for the inspection and maintenance of electrical grid assets, namely insulators and conductors, through its revised guide IEEE-1808-2024 [2].

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Automating visual inspections improves efficiency and consistency by addressing the limitations of manual inspections. Manual checks require skilled workers, and are labor-intensive and time-consuming. Human inspectors inconsistent judgments, promoted by fatigue and cognitive bias, can lead to errors and variability [3]. While automated systems help reduce costs and improve consistency, they come with their own challenges [4], such as, development and maintenance costs, integration with existing systems efforts, or service failure handling overload. Additionally, achieving performance that exceeds human capabilities remains a challenge.

Computer vision generalization capabilities have been lately strengthened by deep learning models. Deep Neural Networks (NN) are surprisingly robust to significant label noise, especially with large datasets. However, given the same amount of training data, clean labels consistently lead to better performance than noisy labels, highlighting that noise in annotations limits the model fitting process [5]. Most recent publications have been employing a model-centric approach, either improving the NN architecture or the learning algorithm [6]. However, the importance of data has been underestimated, resulting in advancements that are not aligned with industry expectations.

Deep learning pioneer Andrew Ng [7], along with his research team and followers, is promoting the data-centric AI movement by sharing its vision of achieving progress not by modifying neural network architectures but by improving data quality. He states that, in industries with limited datasets, the focus should shift from big data to high-quality data, since even a modest collection of meticulously curated examples with consistent and accurate annotations can effectively guide neural networks toward achieving desired learning objectives [8].

Meanwhile, to advance and refine both the conceptual and pragmatic definitions of data quality in the context of supervised deep learning for computer vision, further studies—especially those applying a scientific methodology aligned with the expected outcomes—are essential.

A. Objectives and Contributions

This work analyses the impact of annotation quality and detail level on model performance, evaluating how varying degrees

of annotation precision affect the accuracy and effectiveness of a State-Of-The-Art (SOTA) Deep Vision model. Rather than validating architectural or algorithmic changes to the model, the study uses a data-centric approach, iteratively improving the data quality used to fine-tune several pre-trained variants (small, medium and large) of the 11th YOLO version [9].

The dataset annotations' quality interpretation, adopted in this work, is defined in three dimensions: labeling consistency, spatial granularity, and categorical definition.

- **Labelling Consistency**: Noise mitigation achieved through a strict labeling process. Ensuring quality control policies remain consistent throughout the labeling workflow, require objective rules to guide annotators in ambiguous situations.
- **Spatial Granularity**: Geometric detail level of annotated objects. Specifically, in the conducted experiments, this varies from large bounding boxes to closely fitted polygons.
- **Categorical Definition**: Number of classes considered. The lowest level includes only defective classes, while the highest level combines both defective and functional assets.

The report of the research experiments carried out, present performance results across the three distinct dimensions of data quality, evaluated at varying levels of model complexity and different cross-validation folds, confirming that improving data quality reduces the need for large data quantities.

II. Related Work

The study of automated inspections of electrical grid assets has garnered significant attention over the last decade [1], with numerous approaches exploring model-centric methodologies, including experiments using advanced Multimodal Large Language Models (MLLMs) [10].

The Insulator Defect Image Dataset (IDID) [11] has been cited in over 40 works according to a Google Scholar search conducted at the time of this review. Its high resolution and close-up views of real operational environments make it one of the best publicly available datasets for HV insulators. Since the IDID [11] was shared in a competition context, the test set annotations were kept private. As a result, the test set contains 88 unlabeled images, while the training dataset includes 1,600 annotated images across 4 classes: String Insulator, Good Disc, Broken Disc, and Flashover Shell. However, only 400 images are unique, with the remaining images generated through offline mirroring augmentation, which may require additional care in how the validation of the trained models is performed.

These dataset characteristics and others were properly identified in the study conducted by Fernando Lopes *et al.* [12], leading them to proper careful selection of the images in the validation set. They fine-tuned two SOTA models using 3 subclasses (Good, Broken, and Flashover), achieving validation mAP50 scores of 87.9% and 87.2% with YOLOv8s and Faster R-CNN X101-FPN, respectively. To address the absence of test annotations and quantitative metrics, the authors conducted a qualitative analysis of the prediction results, providing a detailed description of the models' generalization capabilities.

Zheng *et al.* [13] utilized the images from IDID [11] but created a new set of annotations [14] using different implicit rules. Instead of using bounding boxes enclosing the entire asset, they annotated defects present in individual insulator discs, which led to discarding the "good" class. They compared the performance of various SOTA object detectors against their proposed improved YOLOv7 model. The ablation study revealed performance improvements for every component, achieving a mAP50 of 93.8%. However, if the validation split was randomly defined, predictions would be made on mirrored images of those used in training. In practice, such metrics may indicate overfitting capabilities rather than true generalization.

Using the same images, but with different labeling strategies, result in similar yet distinct datasets [11], [14], [15], which might contribute to model performance variations. However, the lack of individual testing experiments prevents distinguishing beneficial labeling rules from harmful ones.

An implicit labeling rule derived from image observations in earlier work [12] is that some "good" insulators, when overlapping with other defects, are not annotated. However, this rule is not consistently applied across all dataset annotations.

The implicit labeling strategy [13] of reducing the bounding box size to fit only the defects could be further improved by implementing polygon' annotations. Although this rule may improve labeling consistency by excluding the "good" class to resolve the overlapping of horizontal bounding boxes, it could also hinder performance by introducing inherent costs.

Intuitively, including the "good" insulator disc class might encourage the neural network to encode disc insulators into feature maps, which could potentially share features with the "broken" disc and "flashover" shell classes. Additionally, this increases the number of instances available for training.

However, from a maintenance perspective, detecting the existence of good insulators has low value, given its high probability of occurrence. The more valuable information lies in detecting rare classes, namely the discussed defects.

Ultralytics' suggestions [16] for dataset creation emphasizes key practices for ensuring high-quality annotations:

- **Label Consistency**: All instances of every class must be labeled over all images in the dataset. Partial labeling might lead to unreliable model performance.
- **Label Accuracy**: Labels must tightly enclose each object, with no gaps between the object and its bounding box.
- **Label Granularity**: The level of detail should align with the task. Polygons should be used for irregular objects, as they offer better accuracy than bounding boxes.
- **Image Variety**: The dataset must represent the deployed environment, capturing images from different times, seasons, weather conditions, lighting, angles, and sources.
- **Background**: About 1-10% of background images (without objects) should be included to reduce False Positives (FP).
- **Dataset Balance**: Bias should be avoided by ensuring a diverse and balanced training set, while over-labeling can promote excessive focus on easy or common classes.
- **Label Verification**: Labels should be verified regularly, and feedback loops should be used to minimize errors. Multiple annotators should be involved to ensure consistency.

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III. Methodology

The experiments in this study were carefully designed to evaluate the impact of various annotation strategies. This resulted in a comprehensive set of experiments, detailed in Table I, that combine various ultra-hyperparameters, ranging from labeling strategy and cross-validation dataset splitting to model complexity. To account for the inherent randomness in the model fitting process, multiple experiments were conducted. Cross-validation was employed to assess the influence of image selection in the training split, while the effect of model complexity was analyzed using different variants of SOTA models.

TABLE I
EXPERIMENTS DIMENSIONS AND DOMAINS CONFIGURATIONS.

Experiment Impact		**Consistency**	**Spatial**	**Categorical**
Dimension	**Domain**	**Range / Count**		
Quality Control	$[basic, ref, careful]$	[1 : 3] **3**	[3] **1**	[2] **1**
Object Shape	$[box_L, box_S, polygon]$	[2] **1**	[1 : 3] **3**	[1] **1**
Classes Discarded	$[none, string, good]$	[2] **1**	[1] **1**	[1 : 3] **3**
X-Val Fold	$[1, 2, 3, 4]$	[1 : 4] **4**	[1 : 4] **4**	[1 : 4] **4**
Model Size	$[small, medium, large]$	[1 : 3] **3**	[1 : 3] **3**	[1 : 3] **3**
Total Count	**108**	**36**	**36**	**36**

Annotation Consistency experiments evaluate the impact of varying the level of Quality Control (QC), ranging from *basic* to *careful*. The *basic* QC policy level represents labeling only obvious defects, while the *careful* QC policy level includes challenging or subtle defects. The *reference* [14] was used as a central threshold for the QC level. Table II provides examples of annotation quality levels and instance counts distribution.

Spatial Granularity influence is assessed in the object shape dimension with a domain ranging from box_L to $polygon$. The box_L level represents large bounding boxes encompassing the entire defective disc. At the other extreme, the $polygon$ level tightly fits the object shape. The small box box_S was obtained by inscribing the respective polygons. Table III presents a visual sample and instance counts for each level.

Categorical Definition assessment was conducted in the classes discarded dimension, with a domain ranging from $none$ to $good$, which includes the 4 original classes (string, good, broken, flash) [11], and the 2 defective classes respectively. An intermediate level includes 3 classes (good, broken, flash), excluding only $string$ insulators. Each variation is illustrated in Table IV, which also provides instance distributions plots.

The Cross-Fold dimension resulted from splitting the existing 400 unique images, ordered by file size, into 4 folds, although the relative object sizes in the images do not necessarily follow the same ordering. The model complexity dimension includes YOLO v11 small, medium, and large variants [9], while all remaining hyperparameters were kept constant. Each experiment's domain spans 3 levels, producing 36 trained model variations (3 levels $\times$ 4 folds $\times$ 3 model sz), resulting in 108 trained models.

TABLE II
METHODOLOGY DESCRIPTION OF ANNOTATIONS CONSISTENCY

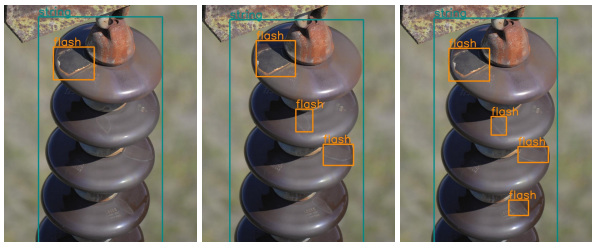
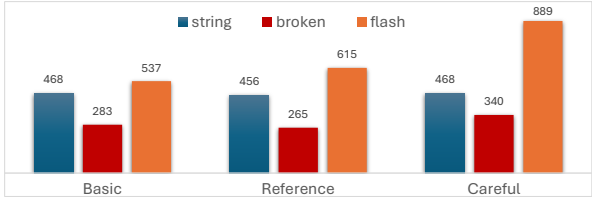
Quality Control	**Basic**	**Reference [14]**	**Careful**
Visual Sample			
Instance Count by Class			

TABLE III
METHODOLOGY DESCRIPTION OF SPATIAL GRANULARITY

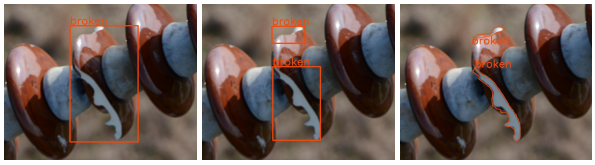
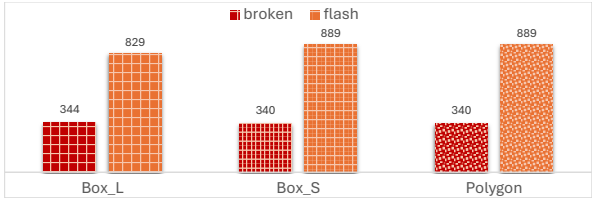
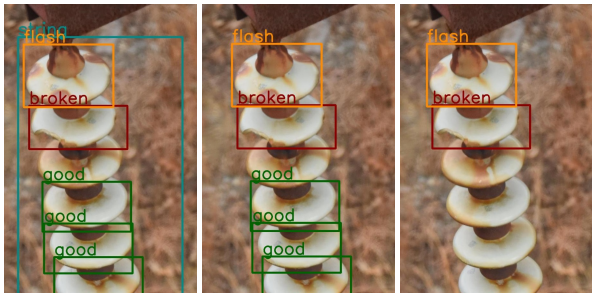
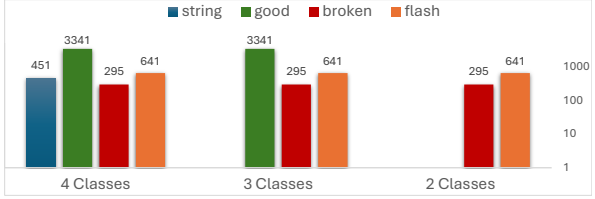
Object Shape	**Large Box**	**Small Box**	**Polygon**
Visual Sample			
Instance Count by Class			

TABLE IV
METHODOLOGY DESCRIPTION OF CATEGORICAL DEFINITION

Classes Droped	**None [11]**	**String**	**String Good**
Visual Sample			
Instance Count by Class			

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IV. Results

To explore how annotation consistency, spatial granularity, and categorical definition affect model performance, this section presents the comprehensive experimental computed results in a compact format, using custom and flexible plotting functions that combine multiple dimensions to effectively distinguish between stochastic variance and deterministic trends.

For consistency, all 108 trained models were evaluated on a separate test set containing 88 images, originally provided by EPRI [11], and annotated by the authors following a careful QC policy. The metrics presented were computed using the ground truth labels and the predictions of the respective trained models. The Average Precision at IoU = 50% (AP50) was chosen as the primary evaluation metric, as it provides a robust measure of object detection accuracy while balancing precision and recall at a commonly used Intersection over Union threshold [17].

From an inspection perspective, the focus lies on the model's ability to detect defective instances. Therefore, in the categorical definition experiments, the test set includes only defective classes (*broken* and *flash*), even though some models were trained with additional functional asset classes.

The average inference time per image ranges from 11.9 ms to 35.4 ms, for the small and large models, respectively, on a system with an AMD Ryzen 9 5900 12-Core, 24-Thread CPU (2.2–4.95 GHz), 64(4x16) GB of RAM DDR4-2400 CL17, and an NVIDIA RTX 4070 Ti GPU (12 GB), running CUDA Driver 12.4, on Kubuntu 22.04, with Conda v24.9.2, Python 3.11.10, CUDA Toolkit 11.5, PyTorch v2.5.1, and Ultralytics v8.3.31.

A. Annotation Consistency

The impact of annotation consistency as a quality control threshold in the dataset labeling procedure was evaluated by analyzing and plotting the 36 respective model' predictions across several standard metrics. While all variations showed high precision, only the *careful* labeling achieved high recall, which is also reflected in AP50, presented at Figure 1, where the cross-validation folds results were averaged to reduce scatter plot clutter and facilitate trend observation.

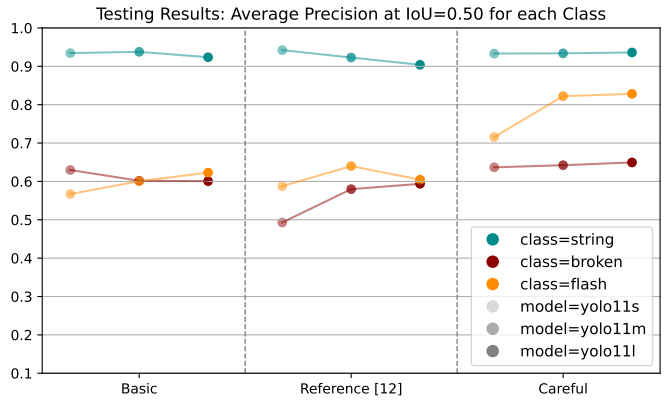


Fig. 1. Scatter plot of AP50 values. The horizontal axis represents labeling QC policies. Classes are represented using color. Model sizes are reflected by color alpha levels and also in a horizontal sub-axis. X-Val folds combinations are reduced to their arithmetic mean.

B. Spatial Granularity

To evaluate the impact of spatial granularity as an object shape fitness level in the dataset labeling procedure, predictions from the 36 respective models in this experiment were analyzed and plotted across several standard metrics. Despite all variations exhibiting similar Recall, a trend between labeling strategies is revealed in Precision. Increasing the IoU threshold in AP impacts box annotations more significantly than polygons. To improve the clarity of observed trends and reduce scatter plot clutter, results were averaged across the model size dimension. These averaged Precision results are presented in Figure 2.

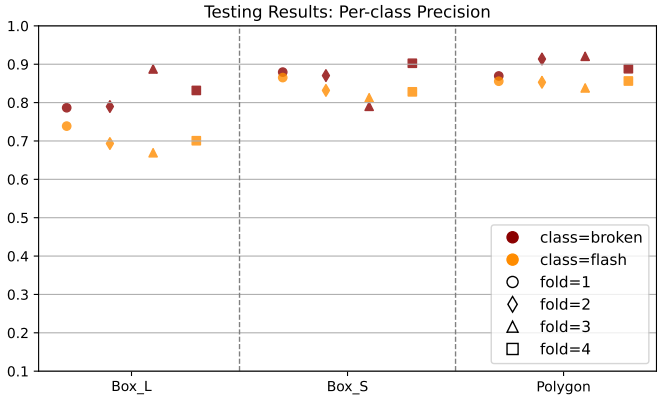


Fig. 2. Scatter plot of Precision values. The horizontal axis represents the object shape considered. Classes are represented using color. X-Val folds are represented using markers and also in a horizontal sub-axis. Performances values were averaged over all model sizes.

C. Categorical Definition

To evaluate the impact of categorical definition, specifically the number of classes considered in the dataset labeling procedure, the performance, of the 36 models from this experiment, were analyzed and plotted across several standard metrics. To enhance trend observation and minimize scatter plot clutter, the performance was averaged over all classes. Figure 3 presents the mean Average Precision at IoU = 50% (mAP50).

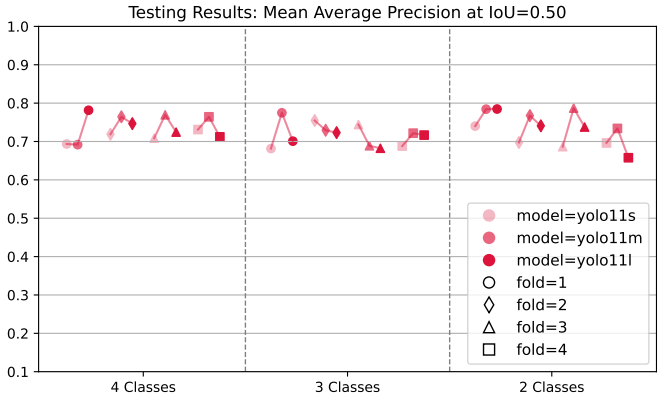


Fig. 3. Scatter plot of mAP50 values. The horizontal axis represents the number of classes included. Model sizes are reflected by color alpha levels and also in a horizontal sub-axis. X-Val folds are represented using markers. Performance was averaged over all classes.

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V. Discussion

Larger models have a greater capacity for learning complex patterns, making them suitable for tasks requiring detailed feature extraction. However, their tendency to memorize noise or irrelevant details makes them prone to overfitting, especially when datasets contain mislabeled data. This helps explain why larger model variants do not always outperform smaller ones.

The results of the annotation consistency experiments indicate marginal variance for the *string* class, since it is a functional asset where the QC threshold is not applicable. It is also notable that the poor results for the *broken* class (AP50=49.3%) occur in the reference dataset [14], which aligns with its lower instance count (265). The standout data points are undoubtedly for the *flash* class, where the highest results (AP50=82.82%) also reflect the larger number of instances (889) annotated using a *careful* labeling policy. This conclusion confirms the intuition that that overlooking annotation procedures, which leads to mislabeled data, limits the model's learning potential.

The spatial granularity experiments reveal an increasing trend in model performance as the boundaries of labeled objects align more closely with their defective visual features, underscoring the importance of high-quality annotation procedures.

The categorical definition experiments exhibit considerable variance across both model and X-Val fold dimensions. However, no significant trends were observed to establish a correlation between the labeling strategy, specifically the inclusion of additional functional classes, and performance. This suggests that the presence of annotated functional assets does not necessarily improve the detection of defective ones.

VI. Conclusion

This study evaluated the impact of various factors on model performance in object detection. Key dimensions such as annotation consistency, spatial granularity, and categorical definition were explored to understand their effects on the performance of trained models. Empirical results identified how different QC policies and labeling strategies influenced model performance.

The results suggest that annotation consistency and the granularity of object representation influences significantly model accuracy. Furthermore, while adding functional asset classes to the dataset did not significantly improve performance, variations in the QC policy and the number of instances annotated in the test set had a notable impact on prediction outcomes.

High scores during development can overstate a model's effectiveness, even when obtained from a well-split dataset. For instance, YOLO11-large achieved 94.8% mAP50 on the validation set with all four classes from the original annotations [11]. However, this inflated performance might have limited value for inspection tasks, since it largely reflects the detection of frequent non-defective classes. On a test set restricted to the two defective classes (*broken* and *flash*), the mAP50 drops to 78.1%, offering a more realistic estimate of performance in operational environments, where the goal is to detect rare anomalies.

This underscores the need for rigorous testing with similar yet distinct datasets and for aligning evaluation criteria with inspection objectives. A key insight from this study is that such reliability ultimately depends on data quality, particularly the annotation procedures applied. The findings therefore reinforce the importance of careful QC practices to minimize noise and reduce the risk of overfitting in model predictions.

Future work could explore contrastive learning loss functions [18], to leverage the inherent differences between defective and functional categories more effectively. Additionally, enriching the dataset with more diverse examples of defective instances and further investigation into the impact of annotation granularity could yield more robust results.

VII. Acknowledgements

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